# Area and Perimeter: Mixed Practice

**Name** _____ **Date** _____ Calculator allowed **Version 1**

**KEY WORDS** area · perimeter · formula · square units · hectare · circumference

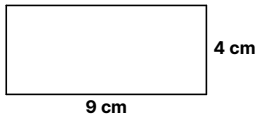
## A Choose the formula ●○○

ANSWER ON THE SHEET

**Remember:** Name the shape first, then write its area formula before you put the numbers in.

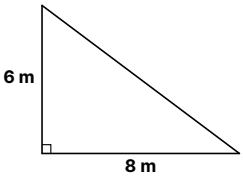
**1.** Find the area of each shape. Give circle answers to 1 decimal place.

**a)**



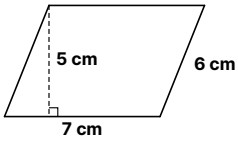

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**b)**



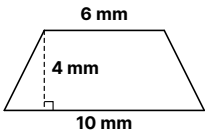

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**c)**



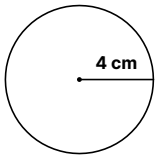

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**d)**



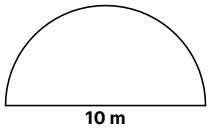

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**e)**




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**f)**




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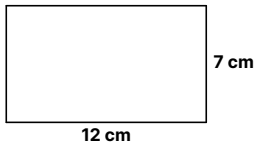
## B Perimeter ●●○

ANSWER ON THE SHEET

**Remember:** Perimeter is the distance around the outside: add every edge. For a curved edge use the circumference formula, and do not forget the straight edges of a semicircle or quarter circle.

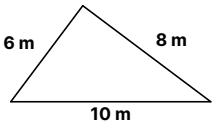
**2.** Find the perimeter of each shape. Give answers with $\pi$ to 1 decimal place.

**a)**



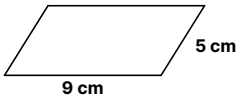

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**b)**



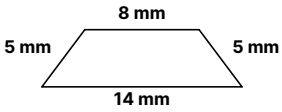

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**c)**



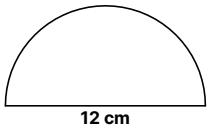

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**d)**



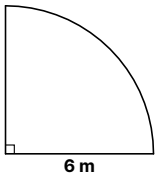

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**e)**




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**f)**




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## C Working backwards ●●○ WORK IN YOUR BOOK

**Remember:** Write the formula, put in what you know, and undo the operations one at a time.

**3.** Find the missing length.

**a)** A square has an area of 144 cm². Find its side length.

**b)** A rectangle has an area of 84 m² and a length of 12 m. Find its width.

**c)** A triangle has an area of 36 cm² and a base of 9 cm. Find its height.

**d)** A circle has a circumference of 50 cm. Find its radius, to 1 decimal place.

**e)** A circle has an area of 200 m². Find its radius, to 1 decimal place.

**f)** A trapezium has parallel sides of 9 cm and 6 cm and an area of 60 cm². Find its height.

## D Units of area ●●○ WORK IN YOUR BOOK

**Remember:** 1 cm² = 100 mm², 1 m² = 10 000 cm², 1 hectare (ha) = 10 000 m², 1 km² = 100 ha. A length unit is multiplied by 10 or 100, so the area unit is multiplied by 10² or 100².

**4.** Change the units.

**a)** Change 3.5 cm² to mm².

**b)** Change 2.4 m² to cm².

**c)** Change 7 ha to m².

**d)** Change 850 mm² to cm².

**e)** Change 45,000 cm² to m².

**f)** Change 3.2 km² to ha.

## E Problems ●●● WORK IN YOUR BOOK

**5.** A rectangular section is 18 m by 12 m.

**a)** Fencing costs $24 per metre. Find the cost of fencing all four sides.

**b)** Lawn seed covers 1 m² for $7.5. Find the cost of seeding the whole section.

**6.** A round trampoline has a diameter of 3.6 m.

**a)** Find the area of the jumping mat, to 2 decimal places.

**b)** Safety padding goes right around the edge and costs $85 per metre. Find the cost of the padding, to the nearest dollar.

**7.** A triangular sail has a base of 4.5 m, a height of 6 m and a longest side of 7.5 m. Rope is sewn along all three edges. The third edge is as long as the height. How much rope is needed?

**8.** A paddock is a rectangle 400 m by 350 m. Find its area in hectares.

## ★ Extension FOR EARLY FINISHERS

**9.** A rectangle has a perimeter of 34 cm and an area of 60 cm². Find its length and width.

**10.** A square and a circle each have a perimeter of 40 cm. How much bigger is the area of the circle than the area of the square? Give the answer to 1 decimal place.



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# Area and Perimeter: Mixed Practice

**Name** _____ **Date** _____ Calculator allowed **Version 2**

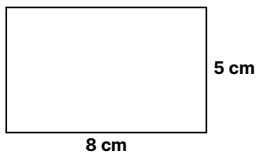
**KEY WORDS** area · perimeter · formula · square units · hectare · circumference

## A Choose the formula ●○○ ANSWER ON THE SHEET

**Remember:** Name the shape first, then write its area formula before you put the numbers in.

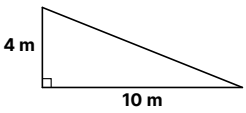
**1.** Find the area of each shape. Give circle answers to 1 decimal place.

**a)**



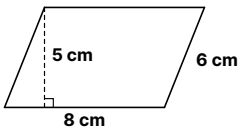

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**b)**



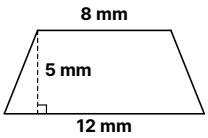

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**c)**



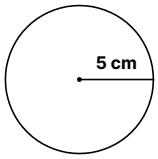

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**d)**



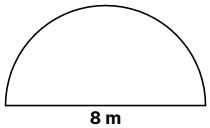

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**e)**




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**f)**



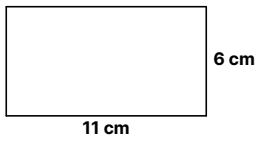

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## B Perimeter ●●○ ANSWER ON THE SHEET

**Remember:** Perimeter is the distance around the outside: add every edge. For a curved edge use the circumference formula, and do not forget the straight edges of a semicircle or quarter circle.

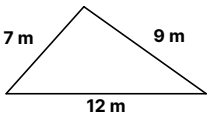
**2.** Find the perimeter of each shape. Give answers with $\pi$ to 1 decimal place.

**a)**



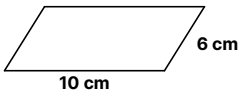

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**b)**



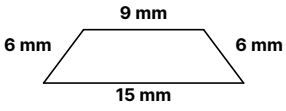

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**c)**



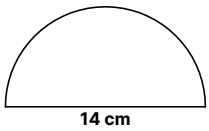

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**d)**



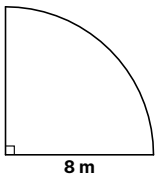

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**e)**




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**f)**




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## C Working backwards ●●○

WORK IN YOUR BOOK

**Remember:** Write the formula, put in what you know, and undo the operations one at a time.

**3.** Find the missing length.

**a)** A square has an area of 169 cm². Find its side length.

**b)** A rectangle has an area of 96 m² and a length of 12 m. Find its width.

**c)** A triangle has an area of 40 cm² and a base of 8 cm. Find its height.

**d)** A circle has a circumference of 60 cm. Find its radius, to 1 decimal place.

**e)** A circle has an area of 250 m². Find its radius, to 1 decimal place.

**f)** A trapezium has parallel sides of 10 cm and 4 cm and an area of 70 cm². Find its height.

## D Units of area ●●○

WORK IN YOUR BOOK

**Remember:** 1 cm² = 100 mm², 1 m² = 10 000 cm², 1 hectare (ha) = 10 000 m², 1 km² = 100 ha. A length unit is multiplied by 10 or 100, so the area unit is multiplied by 10² or 100².

**4.** Change the units.

**a)** Change 4.5 cm² to mm².

**b)** Change 1.8 m² to cm².

**c)** Change 6 ha to m².

**d)** Change 640 mm² to cm².

**e)** Change 72,000 cm² to m².

**f)** Change 2.6 km² to ha.

## E Problems ●●●

WORK IN YOUR BOOK

**5.** A rectangular section is 20 m by 14 m.

**a)** Fencing costs $26 per metre. Find the cost of fencing all four sides.

**b)** Lawn seed covers 1 m² for $8. Find the cost of seeding the whole section.

**6.** A round trampoline has a diameter of 4.2 m.

**a)** Find the area of the jumping mat, to 2 decimal places.

**b)** Safety padding goes right around the edge and costs $90 per metre. Find the cost of the padding, to the nearest dollar.

**7.** A triangular sail has a base of 5.2 m, a height of 6 m and a longest side of 7.9 m. Rope is sewn along all three edges. The third edge is as long as the height. How much rope is needed?

**8.** A paddock is a rectangle 450 m by 300 m. Find its area in hectares.

## ★ Extension

FOR EARLY FINISHERS

**9.** A rectangle has a perimeter of 22 cm and an area of 24 cm². Find its length and width.

**10.** A square and a circle each have a perimeter of 36 cm. How much bigger is the area of the circle than the area of the square? Give the answer to 1 decimal place.



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